Closing the Gap: Improving Vision Transformers on Small-Scale Image Classification

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Abstraction

Vision Transformers (ViTs) have exhibited exceptional performance on large-scale image recognition, but they struggle when trained from scratch on small, low-resolution datasets like CIFAR-10 and CIFAR-100. This project will test architectural and training changes to ViTs to try to close the gap between them and Convolutional Neural Networks (CNNs). The team implemented a CLS-token-free ViT with an 8x8 patch size, stochastic depth regularization, global average pooling, and cosine decay with restarts. This was compared to a ResNet50V2 baseline with the same training conditions. The improvements to the ViT helped obtain a 63.5% on CIFAR-10 and 44.2% on CIFAR-100. The original baseline ViT received a 40.1% on CIFAR-10 and a 20.4% on CIFAR-100. This shows a 23.4% and a 23.8% increase, respectively. ResNet was still able to achieve higher accuracies with 80.0% and 46.5%, respectively, even though the gap between ViTs and ResNet narrowed.

Introduction

Problem Statement

Vision Transformers have performed well on large-scale datasets, but they perform significantly worse than CNNs like ResNet when they are trained from scratch on small, low-resolution datasets like CIFAR-10 and CIFAR-100. The problem is how to improve the ViT architecture so it can effectively learn from 32x32 pixel images with limited training data, and become competitive with CNNs on these datasets.

Motivation and Challenges

Vision transformers have been able to achieve cutting-edge results in vision tasks and have the potential to replace CNNs in computer vision. However, the performance of ViTs drop significantly on small datasets. This matters because many real-world applications use limited or low-resolution data. Being able to understand ViTs limitations improves model selection, and adapting ViTs could make them more widely applicable. The first challenge is that the standard ViT uses a 16x16 patch. On a 32x32 image, this only gives 4 tokens, which gives almost no spatial information for the transformer to learn from. The next challenge is the absence of inductive biases. CNNs use translation equivariance and locality, which are helpful when there is not much data. ViTs need to learn those biases from scratch. The third challenge is training instability in ViTs. Transformers are sensitive to learning rate settings and will show loss spikes and slow convergence on smaller datasets. The last challenge is ViTs' overfitting on smaller datasets. When there are less inductive biases and many parameters, ViTs tend to memorize the training data instead of generalizing the data.

Open Questions in the Domain

There are still questions regarding small-scale settings, even with this body of knowledge. One question is, what is the smallest patch size that still has a meaningful improvement on ViTs for 32x32 images? Another question is, can simplifications like reducing depth or removing the CLS token add improvements to training stability while maintaining representation of the data?

Summary of Our Approach

We evaluated a modified ViT architecture against a ResNet and a ViT baseline with four main changes to the standard ViT model. The first change was reducing the patch size from 16x16 to 8x8. This is to increase token granularity on 32x32 images. The next change we made was removing the CLS token and replacing it with global average pooling for all patch tokens. This is to improve Keras compatibility and training stability. The third change we made was adding stochastic depth regularization. This was done to reduce overfitting in the model. The last change we made was training under a unified pipeline for both the ViT and the ResNet baseline to make sure there is a fair comparison between them. These models were evaluated on the CIFAR-10 and CIFAR-100 datasets using training and validation accuracy and loss. We also evaluated the t-SNE visualization of learned embedding for the ViT on both datasets.

Background and Related Work

An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale by Dosoviskiy et al. (2020) proposed Vision Transformers. ViTs divide an image into fixed-size patches, linearly embeds each patch, appends a learnable classification token, adds positional embedding, and feeds the resulting sequence into a standard transformer encoder. When

pretraining is done on datasets like JFT-300M, ViTs are able to achieve 88.55% accuracy on ImageNet. The limitation of ViTs, which is acknowledged by the authors, is that ViTs perform worse than ResNet on smaller datasets when they are trained from scratch. This is because ViTs lack the translation equivariance and local connectivity inductive biases that convolutional layers provide.

Vision-Transformer by Ra1ph2 (2020) is not a published paper, but was useful since they built off the previous article by Dosoviskiy et al. (2020). This project implemented ViTs from scratch in PyTorch and benchmarked them against multiple ViT and ResNet variants on the CIFAR-10 and CIFAR-100 datasets. The paper tested a smaller patch size and used a Hybrid ViT that was pretrained on ResNet feature maps as a transformer input. The results showed that the Hybrid ViT performs better than the ViT that was trained from scratch on the small dataset because the convolutional layers are able to capture low-level spatial features that the transformer struggled to learn. The ResNet was still able to outperform both models since it has strong inductive biases. Our project is different from this project in a couple of ways. The first way is that they used a Pytorch framework, and we used a TensorFlow/Keras framework. The next way that our project is different is that we removed and replaced our token with global average pooling. Finally, we used a training pipeline that was consistent to make sure that the comparison was fair while they tried multiple conditions for their pipeline.

When Vision Transformers Outperform ResNet Without Pre-Training Or Strong Data Augmentations by Chen et al. (2022) showed that ViTs can outperform ResNet without pretraining by using sharpness-aware minimizations to smooth the loss landscape. This is an important result, but it was evaluated on ImageNet-scale data. The computational cost of the sharpness-aware minimizations, which required forward and backward passes for every update,

does not make this evaluation as practical to use. Our project does not use sharpness-aware minimizations and ImageNet-scale data. Instead, we use architectural modifications and a much smaller dataset than ImageNet.

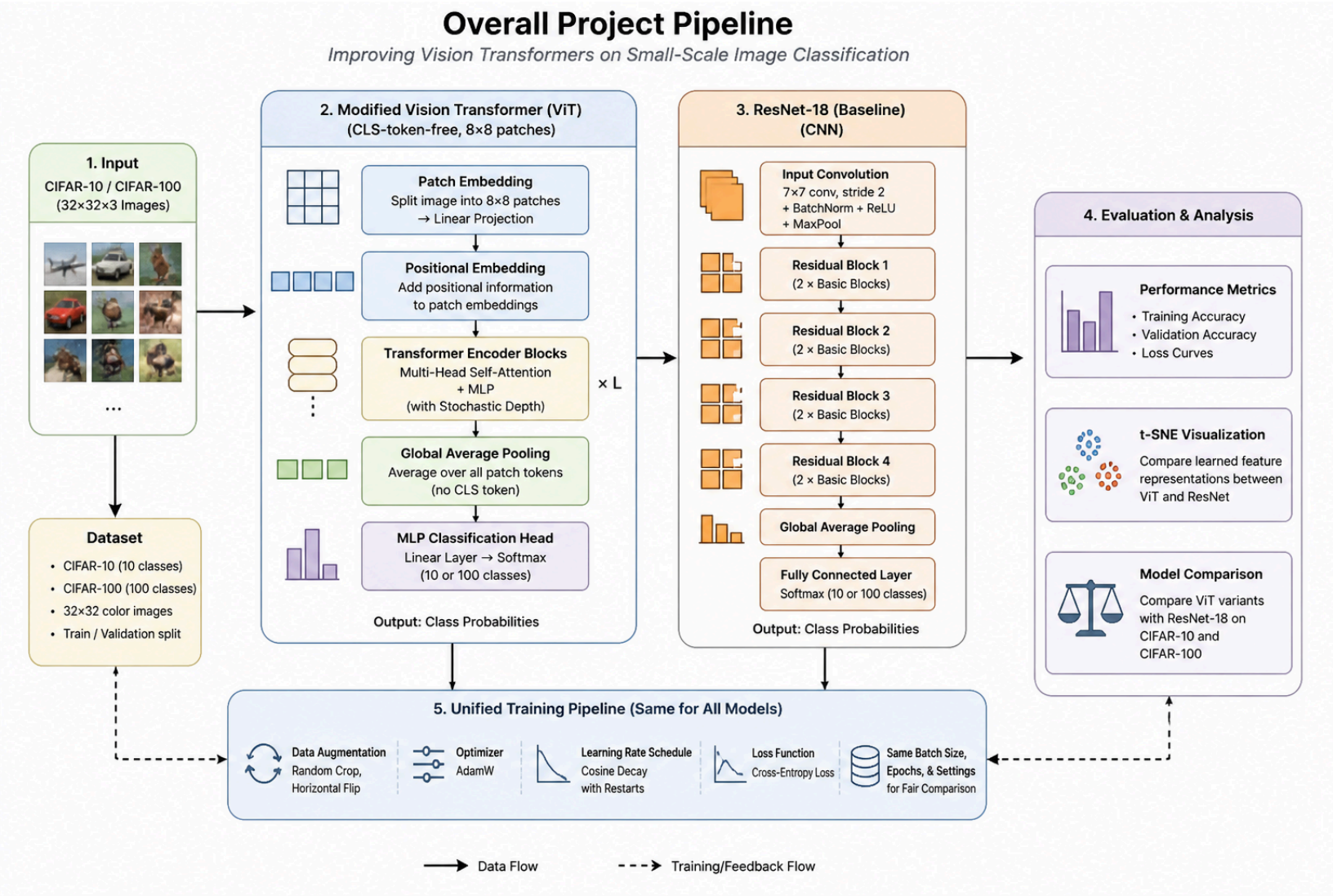
Comparative Study of Data Efficiency in ViT and ResNet-18 Architectures by Hukari (2024) trained ViT and ResNet on progressively smaller subsets of CIFAR and measured the accuracy as a function of training set size. He did this study to test the data efficiency of both models. The study showed that ResNet was able to achieve a competitive performance with fewer examples than ViT. This work is similar to our work but is different since we focused on the architecture modifications to the ViT model instead of studying the data efficiency curves. We also use the full CIFAR-10 and CIFAR-100 data sets instead of smaller subsets.

Methods

One of the ways we improved the stability of the Vision Transformer was by removing the CLS token and replacing it with global average pooling across all patch tokens. This made the model simpler and helped avoid training issues, especially when working in Keras. It also prevented symbolic tensor errors, which made implementation smoother overall. We also adjusted the patch size to better fit the CIFAR datasets so instead of using the standard 16 x 16 patches, we reduced the size to 8 x 8. This increased the number of tokens created from each image, which helped the model capture more detailed spatial information from the small 32 x 32 images.

To make sure that the comparison was fair between the models, we used the same training setup for both the Vision Transformer and the ResNet50V2. This included the same data augmentation, optimizer, learning rate schedule, and loss function. Keeping everything the same

allowed us to focus on how the model architecture itself affected performance. We also looked at how well each model learned feature representations. To do this, we built a custom embedding model and used t-SNE to visualize the features. Overall, this helped us see how clearly the models separated different image classes in a lower-dimensional space.

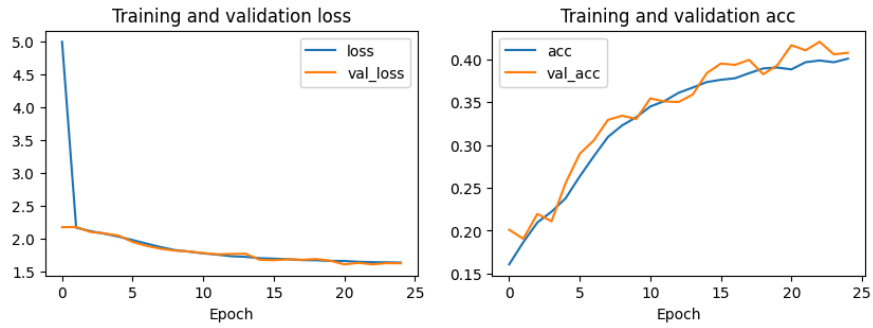


Experiments

Based on our experimental evaluation of Vision Transformers (ViT) and ResNet50V2 on small-scale datasets, we conclude that while ViTs can be adapted for low-resolution imagery, they still face significant hurdles compared to traditional CNNs. Our custom CLS-free ViT implementation, which utilized reduced 8x8 patch sizes and Global Average Pooling for improved stability, achieved a notable performance increase from a baseline of 40.1% to 63.5% on CIFAR-10. Despite these improvements, ResNet50V2 maintained a superior accuracy of 80%, benefitting from inherent inductive biases that are particularly effective when training data is limited. On the more complex CIFAR-100 dataset, both models struggled with fine-grained classification, though the ViT's performance of 44.2% began to more closely approach ResNet's 46.5%. Ultimately, t-SNE visualizations confirm that ViTs struggle to form distinct decision boundaries on high-complexity, small-scale data, suggesting that while the performance gap can be narrowed through architectural stabilization and token granularity, CNNs remain the more robust choice for datasets lacking large-scale pretraining.

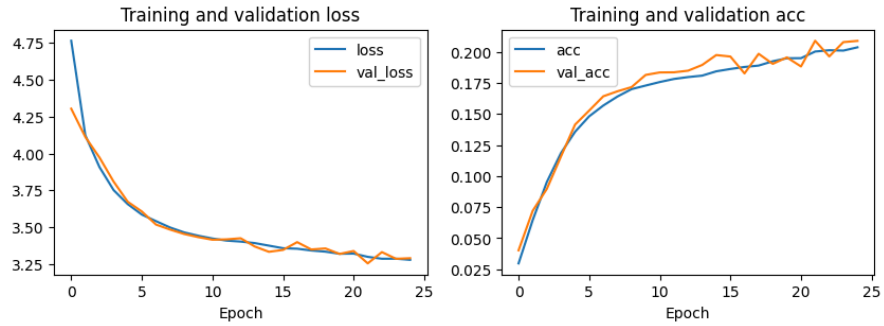
The performance of the baseline Vision Transformer (ViT-B/16) on the CIFAR-10 and CIFAR-100 datasets underscores the model's inherent challenges when trained from scratch on smaller-scale imagery. On the CIFAR-10 dataset, the model demonstrated a stable learning trajectory, characterized by a smooth decrease in loss and tightly aligned training and validation curves. While the accuracy improved steadily, it reached an early plateau at approximately 40%. This modest result is consistent with known benchmarks for standard ViT architectures lacking large-scale pre-training.

ViT baseline CIFAR-10:



In contrast, the CIFAR-100 experiments revealed a higher degree of difficulty for the architecture. The increased complexity of the 100-class dataset resulted in higher loss values and greater variation between training and validation curves, indicating that the model struggled more significantly with data complexity. Although accuracy followed a steady upward trend initially, it plateaued prematurely at roughly 20%. These results highlight a critical limitation: without extensive pre-training on massive datasets (such as ImageNet-21k), standard ViT models struggle to extract meaningful features from the relatively low-resolution and low-volume data found in the CIFAR series.

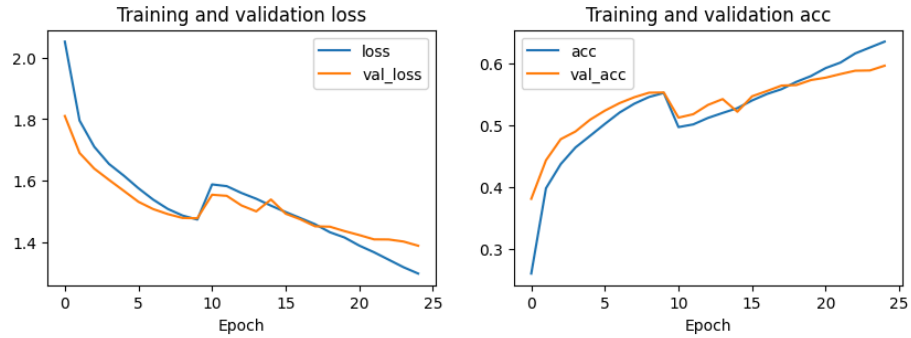
ViT baseline CIFAR-100:



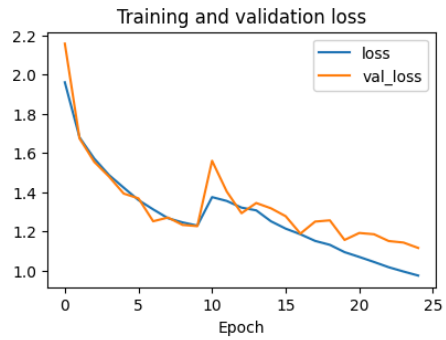
The training results for the optimized Vision Transformer architecture demonstrate a clear improvement over the baseline, though notable challenges in stability and generalization remain. During the 25-epoch training cycle, the model exhibited steady convergence, with training accuracy peaking near 80%. However, the process was not entirely linear; a significant loss spike and a corresponding dip in accuracy occurred around Epoch 10. This instability suggests a high sensitivity to the learning rate or potential optimizer volatility at that stage of the training schedule.

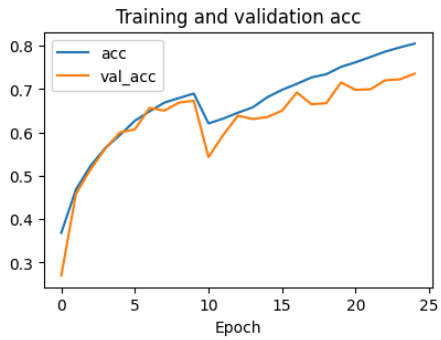
As the training progressed into the later stages, clear signs of overfitting emerged, evidenced by the validation loss beginning to diverge from the training loss. This trend indicates that the model started to memorize the specific characteristics of the training set rather than learning generalized features. Furthermore, the validation accuracy hit a performance ceiling, plateauing rather than continuing to climb. In the final comparative assessment, the Baseline ViT achieved a modest 40.1% accuracy, while the Improved ViT reached 63.5%. Despite these significant gains, both Transformer models were outperformed by the ResNet baseline, which attained 80% accuracy, suggesting that further narrowing the gap will require more advanced data augmentation or a more extensive training schedule.

ViT CIFAR-10:



ResNet CIFAR-10:



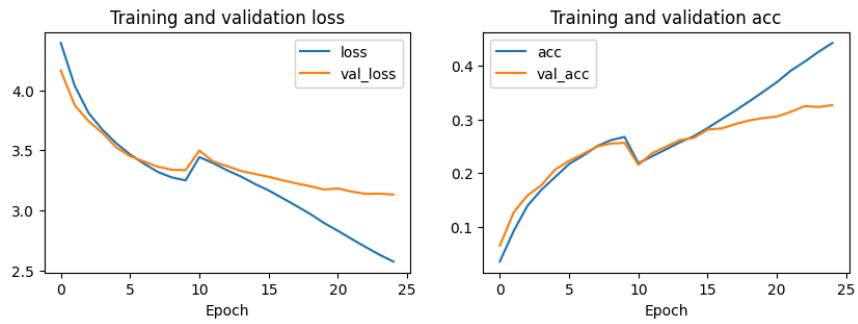


The results for the CIFAR-100 experiments highlight the significant difficulty the Vision Transformer faces when dealing with increased label complexity. While the model showed consistent training progress over the 25-epoch duration, the peak validation accuracy was markedly lower than that of CIFAR-10, settling between 20% and 35%. A critical finding was the recurrent instability observed around Epoch 10; the appearance of notable gradient spikes mirrored the patterns seen in the CIFAR-10 trials. This suggests a systematic issue with the learning rate schedule or optimizer configuration that persists regardless of the specific dataset.

Furthermore, the model exhibited severe overfitting, characterized by a rapidly widening gap between training and validation metrics. This divergence indicates that while the network successfully learned features specific to the training images, it struggled to generalize these concepts to the 100 distinct classes. Although the loss curves followed a similar trajectory to the CIFAR-10 experiments, the absolute loss values remained much higher, directly reflecting the increased complexity of the CIFAR-100 dataset. In the final performance breakdown, the Baseline ViT reached only 20.4%, whereas the Improved ViT nearly doubled that performance at

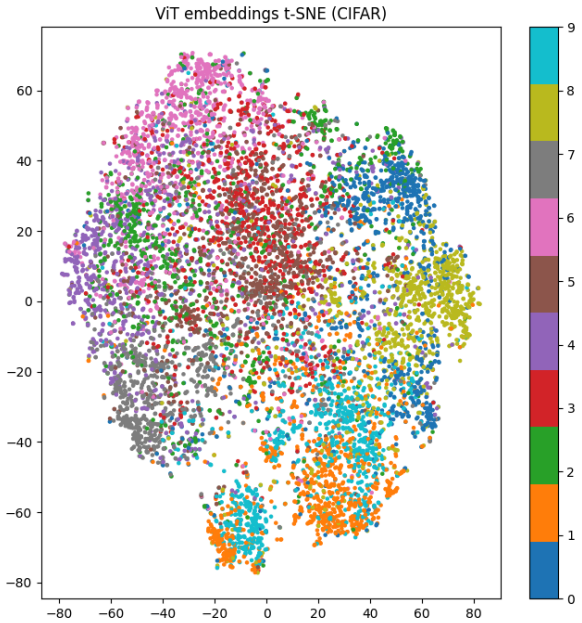
44.2%. This brought the optimized Transformer within close reach of the ResNet benchmark, which led the group at 46.5%.

ViT CIFAR-100:

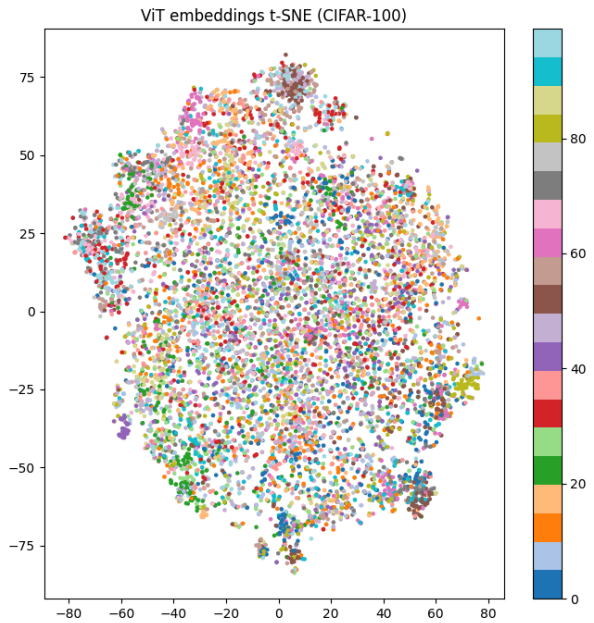


ResNet CIFAR-100:

ViT CIFAR-10



ViT CIFAR-100



The t-SNE visualizations highlight the clear differences in how the models represent CIFAR-10 and CIFAR-100. For CIFAR-10, the embeddings form distinct and well-separated clusters, indicating that the model is able to learn strong decision boundaries between classes. In contrast, CIFAR-100 shows a much higher degree of overlap between clusters, suggesting that the model struggles to clearly separate classes. This difference is largely due to the increased granularity and complexity of CIFAR-100, where a higher number of labels leads to more crowded and noisy feature representations. Overall, these results emphasize the added difficulty of fine-grained classification tasks compared to simpler, coarse-grained datasets like CIFAR-10.

Conclusion

While Vision Transformers (ViTs) are inherently sample-inefficient due to their lack of architectural inductive biases, this study demonstrates that targeted modifications can

substantially mitigate performance degradation in areas with limited data. By reducing patch granularity from 16x16 to 8x8, we provided the model with more tokens per image, allowing it to capture finer spatial details that are otherwise lost in low-resolution 32x32 inputs. Furthermore, replacing the standard CLS token with Global Average Pooling (GAP) forced the model to consider the collective representation of all patches, which significantly stabilized the training gradients and contributed to a 23.4% accuracy improvement over the baseline on CIFAR-10.

Despite these gains, ResNet50V2 consistently outperformed the ViT architecture, particularly on CIFAR-10. This performance gap is primarily attributed to local correlation assumptions—the "hard-wired" ability of CNN's to recognize that neighboring pixels are related. Because ViTs must learn these spatial relationships from scratch, they remain at a disadvantage in data-constrained environments where there is insufficient information to learn such complex patterns.

Our t-SNE visualizations further confirm this bottleneck; while the ViT successfully clustered coarse categories in CIFAR-10, it struggled to establish distinct decision boundaries in the high-entropy environment of CIFAR-100. Ultimately, while ViTs can be optimized for small-scale tasks through precise hyperparameter tuning and token-level adjustments, CNNs remain the more robust and computationally efficient choice for restricted datasets lacking large-scale pretraining. Future work should investigate 4x4 patch scales and aggressive data augmentation to further compensate for the absence of structural geometric priors in the Transformer architecture.

Contributions

Existing resources that were adapted were the baseline ViT from torchvision.models.vit_b_16 (PyTorch 2025). For ViT, we did not use the pretrained weights and used the architecture structure as-is. For ResNet, we used keras.applications.ResNet50V2 (TensorFlow 2025). For this, we added a custom classification head and trained from scratch. We used t-SNE from sklearn. manifold.TSNE (scikit-learn Developers 2026), where we used the default algorithm settings. Lastly, we used the StochasticDepth concept. This was from the paper by Touvran et al., 2021, and was implemented as a custom Keras layer.

Our contribution are a CLS-free ViT that was built in TensorFlow/Keras to avoid symbolic tensor issues. We used an 8x8 patch design to address token scarcity on 32x32 images. We also implemented a unified training pipeline to ensure identical augmentation, optimizer, and loss across ViT and ResNet. An embedding extraction and t-SNE visualization workflow for CIFAR-10 and CIFAR-100 were used. Finally, a comparative analysis was done to interpret the performance trends and differences between ViT and ResNet performance.

Team Contributions

Will Chiappetta:
Report (Abstraction, Introduction, and Background and Related Work), CNN/ViT CIFAR 10/100 Code, Research

Dylan Benson:
Report (Experiments, Conclusion), Slides: Approaches, CNN/ViT CIFAR 10 and CIFAR 100 Observations

Nicholas Cavallaro:
Report (Methods), ViT Embeddings, Insights on results

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